

MAT9004 - Assignment 2

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Ans 1

Fund	Unit Price	Shares	Property	Bonds	Return
A	\$ 3.25	20%	75%	5%	7%
B	\$ 2.50	35%	30%	35%	3%
C	\$ 6.75	45%	50%	5%	6%
D	\$ 4.00	75%	10%	15%	9%

Investment manager is required to have :-
 (1% of 1 million)

50% to shares $\Rightarrow$ 500 (thousand)
 30% to property $\Rightarrow$ 300 (thousand)
 20% to Bonds $\Rightarrow$ 200 (thousand)

a) If a, b, c, d is the amount in 1000's of dollar then the equation becomes :-

1. Shares Exposure equation :-

$$0.2a + 0.35b + 0.45c + 0.75d = 500$$

2. Property Exposure equation :-

$$0.75a + 0.3b + 0.5c + \overset{0.1d}{\cancel{0.01d}} = 300$$

3. Bond Exposure equation

$$0.05a + 0.35b + 0.05c + 0.15d = 200$$

The sum of all investment will be :- $a + b + c + d = 1000$

6) In matrix form :- $Ax = B$

$$\begin{pmatrix} 0.2 & 0.35 & 0.45 & 0.75 \\ 0.75 & 0.3 & 0.5 & 0.10 \\ 0.05 & 0.35 & 0.05 & 0.15 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 500 \\ 300 \\ 200 \\ 1000 \end{pmatrix}$$

$A \quad x \quad B$

In this matrix form we can say the fourth equation
i.e.

$$a + b + c + d = 1000$$

is redundant as :-

When the rest of the 3 equations including these :-
:- Bonds
:- Property
 are solved the total invested amount constraint
 will also be met.

So, we can remove any one of the four equations here.

i.e. :-

$$\begin{bmatrix} 0.2 & 0.35 & 0.45 & 0.75 \\ 0.75 & 0.3 & 0.5 & 0.10 \\ 0.05 & 0.35 & 0.05 & 0.15 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 500 \\ 300 \\ 200 \end{bmatrix}$$

c) Using Gaussian elimination to find unique solutions:-

$$\begin{bmatrix} 0.2 & 0.35 & 0.45 & 0.75 \\ 0.75 & 0.3 & 0.5 & 0.10 \\ 0.05 & 0.35 & 0.05 & 0.15 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 500 \\ 300 \\ 200 \\ 1000 \end{bmatrix}$$

$$R_1 \rightarrow R_1 / 0.2 \rightarrow R_1$$

$$\begin{bmatrix} 1 & 1.75 & 2.25 & 3.75 \\ 0.75 & 0.3 & 0.5 & 0.10 \\ 0.05 & 0.35 & 0.05 & 0.15 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2500 \\ 300 \\ 200 \\ 1000 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 0.75 R_1$$

$$R_3 \rightarrow R_3 - 0.05 R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\begin{bmatrix} 1 & 1.75 & 2.25 & 3.75 \\ 0 & -1.0125 & -1.1875 & -2.7125 \\ 0 & 0.2625 & -0.0625 & -0.0375 \\ 0 & -0.75 & -1.25 & -2.75 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2500 \\ -1575 \\ 75 \\ -1500 \end{bmatrix}$$

$$R_2 \rightarrow R_2 / -1.0125$$

$$\begin{bmatrix} 1 & 1.75 & 2.25 & 3.75 \\ 0 & 1 & \frac{95}{81} & \frac{217}{81} \\ 0 & 0.2625 & -0.0625 & -0.0375 \\ 0 & -0.75 & -1.25 & -2.75 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2500 \\ 14000/9 \\ 75 \\ -1500 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 1.75 R_2$$

$$R_3 \rightarrow R_3 - 0.2625 R_2$$

$$R_4 \rightarrow R_4 + 0.75 R_2$$

$$\begin{bmatrix} 1 & 0 & \frac{16}{81} & -\frac{76}{81} \\ 0 & 1 & \frac{95}{81} & \frac{217}{81} \\ 0 & 0 & -\frac{10}{27} & \frac{20}{27} \\ 0 & 0 & -\frac{10}{27} & -\frac{20}{27} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} -\frac{2000}{9} \\ \frac{14000}{9} \\ -\frac{1000}{3} \\ -\frac{1000}{3} \end{bmatrix}$$

$$R_3 / -\frac{10}{27} \rightarrow R_3$$

$$\begin{bmatrix} 1 & 0 & \frac{16}{81} & -\frac{76}{81} \\ 0 & 1 & \frac{95}{81} & \frac{217}{81} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -\frac{10}{27} & -\frac{20}{27} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} -\frac{2000}{9} \\ \frac{14000}{9} \\ 900 \\ -\frac{1000}{3} \end{bmatrix}$$

$$R_1 \rightarrow R_1 - \frac{16}{81} R_3$$

$$R_2 \rightarrow R_2 - \frac{95}{81} R_3$$

$$R_4 \rightarrow R_4 + \frac{10}{27} R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{4}{3} \\ 0 & 1 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} -400 \\ 500 \\ 900 \\ 0 \end{bmatrix}$$

$\therefore$ Now the solution is :-

i) $a - \frac{4}{3}d = -400$

ii) $6 + \frac{1}{3}d = 500$

iii) $c + 2d = 900$

d) According to the table given :-

$\rightarrow$ We can see the maximum return is from the Fund D
i.e. 9%.

$\rightarrow$ Followed by the Fund A (7%).

Fund C (6%).

Fund B (3%).

Q2 a) As $f(x) = e^{-cx}$

$\therefore$ to find the tangent lines, finding derivatives

$$f'(x) = -ce^{-cx}$$

And As the tangent crosses through the origin the equation of the line will be in:

$$y = mx$$

$\therefore$ at the origin the value of the derivative is the slope of the required tangent

$$\therefore \text{Eqn} \Rightarrow y = -ce^{-cx}x$$

Similarly:-

$$\text{As } g(x) = e^{cx}$$

finding the ~~tan~~ derivative:-

$$g'(x) = ce^{cx}$$

$$\therefore y = mx \text{ where } m = ce^{cx}$$

$$\text{Eqn} \Rightarrow y = ce^{cx}x$$

for the Point A, B

$$\text{For } f(x): y = -ce^{-cx}x$$

$$\text{For } g(x): y = ce^{cx}x$$

26) Finding the tangent vectors v_A and v_B

$$\therefore \text{for } f(x) = e^{(-cx)} \\ f'(x) = -ce^{-cx}$$

$\therefore$ at point A, $x=0$

the slope of the tangent line is $-ce^{(-c \cdot 0)}$

26) for the orthogonal check

$$\boxed{v_A \cdot v_B = 0}$$

~~from~~ for $f(x) \therefore y = -ce^{-cx}$

for $g(x) \therefore y = ce^{cx}$

in vector form $\therefore v_A = [1, -ce^{-cx}]$

$$v_B = [1, ce^{cx}]$$

$$\therefore v_A \cdot v_B = (1)(1) + (-ce^{-cx})(ce^{cx})$$

$$= 1 - c^2 e^{c(b-a)}$$

$\therefore$ for orthogonality $\therefore$

$$c^2 e^{c(b-a)} = 1$$

$$e^{c(b-a)} = \frac{1}{c^2}$$

$$c(b-a) = -2 \ln(c)$$

$$\therefore (b-a) = \frac{-2 \ln(c)}{c}$$

b) ~~The tangents are orthogonal to each other~~

$$\therefore V_A \cdot V_B = 0$$

c) To find the area of the shaded region :-

We ~~can~~ can find the area under $f(x)$ from point A to the $x=0$ and multiply it by 2

As both the axes are symmetrical.

$\therefore$ for Area :-

$$f(x) = e^{-cx} \text{ for } x \leq 0$$

$$\text{eqn of tangent from A :- } y = mx + e^{-cx}$$

$$\text{at point A :- } \frac{y - e^{-ca}}{x - a} = -ce^{-ca}$$

$$\text{As } (x, y) = (0, 0) \text{ is for tangent } \frac{e^{-ca}}{a} = -ce^{-ca}$$

$$\therefore \boxed{a = \frac{1}{c}}$$

$$\therefore y = -ce\left(x + \frac{1}{c}\right) + e$$

$$\therefore \text{Area :- } \int_{-\frac{1}{c}}^0 (f(x) - \text{tangent}) dx$$

$$\Rightarrow \int_{\frac{1}{c}}^0 \left(e^{-cx} - \left[-ce \left(x + \frac{1}{c} \right) + e \right] \right) dx$$

$$\Rightarrow -\frac{1}{c} \left[e^{-cx} \right]_{\frac{1}{c}}^0 - \left[-ce \frac{x^2}{2} + ex + ex \right]_{\frac{1}{c}}^0$$

$$\Rightarrow -\frac{1}{c} [1 - e] - \left[0 + \frac{e}{2c} \right] \Rightarrow -\frac{1}{c} [1 - e] + \left[\frac{e}{2c} \right]$$

$$\Rightarrow -\frac{1}{c} \left[1 - e + \frac{e}{2} \right]$$

$$= \frac{1}{c} \left[\frac{e}{2} - 1 \right]$$

$\therefore$ The total area will be 2 times this value

$$\Rightarrow \underline{\text{Area}} \quad \frac{2}{c} \left[\frac{e}{2} - 1 \right]$$

$$\frac{e}{c} - \frac{2}{c} \quad \frac{1}{c} [e - 2]$$

Q3. Matrix :-

$$A = \frac{1}{5} \begin{bmatrix} 39 & -48 \\ -48 & 11 \end{bmatrix}$$

a) Show that if v is an eigenvector with eigenvalue λ of any matrix B , then v is also an eigenvector of the matrix (cB) for any constant $c \in \mathbb{R}$.

$$\therefore \text{If } BV = \lambda v \text{ (Given) } \quad \text{--- (1)}$$

$\therefore$ using matrix (cB)

$$(cB)v = c(Bv)$$

using (1)

$$(cB)v = (c\lambda)v$$

$\therefore v$ is the eigen vector of cB
where eigen value is $c\lambda$

26) ~~The tangent lines can be:~~

3)

$$A = \frac{1}{5} \begin{pmatrix} 39 & -48 \\ -48 & 11 \end{pmatrix}$$

6) Using $A = CB$

$$C \Rightarrow \frac{1}{5}$$

$\therefore$ for calculating B

$$CB = \lambda I$$

$$\det(B - \lambda I) = 0$$

$$\text{That is: } |B - \lambda I| = \begin{vmatrix} 39 & -48 \\ -48 & 11 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix}$$

$$\therefore (39 - \lambda)(11 - \lambda) - (-48)(-48)$$

$$\Rightarrow 429 - 39\lambda - 11\lambda + \lambda^2 - 2304$$

$$(B - \lambda I) = \lambda^2 - 50\lambda - 1875$$

$$\lambda^2 - 50\lambda - 1875 = 0$$

$$\lambda^2 - 75\lambda + 25\lambda - 1875 = 0$$

$$\lambda(\lambda - 75) + 25(\lambda - 75) = 0$$

$$\therefore \lambda = 75, -25$$

$\therefore$ The eigen values of A can be detected using

$$A = C\lambda$$

$$\lambda_1 = 75 \times \frac{1}{5} = 15$$

$$\lambda_2 = -25 \times \frac{1}{5} = -5$$

c) Using the prev. values of λ_1 & λ_2 i.e 15 & -5

When $\lambda_1 = 15$

$$\therefore \text{for } (A - 15I)x = 0$$

$$A - 15I = \frac{1}{5} \begin{pmatrix} 39 & -48 \\ -48 & 11 \end{pmatrix} - 15 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{-36}{5} & \frac{-48}{5} \\ \frac{-48}{5} & \frac{-64}{5} \end{pmatrix}$$

For eigenvectors we need to solve:-

$$\begin{pmatrix} -36 & -48 \\ -48 & -64 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\therefore \begin{aligned} -36x_1 - 48x_2 &= 0 \\ 48x_1 + 64x_2 &= 0 \end{aligned}$$

$$\therefore \boxed{x_1 = -\frac{4}{3}x_2}$$

$\therefore$ for $\lambda_1 = 15$ the eigenvector $(v_1) = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$

for Eigenvalue $\lambda_2 = -5 \Rightarrow (A + 5I)v = 0$

$$A + 5I = \frac{1}{5} \begin{pmatrix} 39 & -48 \\ -48 & 11 \end{pmatrix} + 5 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{64}{5} & -\frac{48}{5} \\ -\frac{48}{5} & \frac{36}{5} \end{pmatrix}$$

$\therefore$ Solving $\therefore \begin{pmatrix} 64 & -48 \\ -48 & 36 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$64v_1 - 48v_2 = 0$$

$$-48v_1 + 36v_2 = 0$$

Here $\therefore v_1 = \frac{3}{4}v_2$

$\therefore$ for the eigenvalue $\lambda_2 = -5$ eigenvector $v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

d) To check the orthogonality of two vectors

we need to prove the dot product of them to be $= 0$

$$\therefore \underline{v_1 \cdot v_2 = 0}$$

$$v_1 \cdot v_2 \Rightarrow \begin{bmatrix} -4 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} \Rightarrow (-4)(3) + (3)(4) \\ = -12 + 12 \\ = 0$$

$\therefore$ the two eigenvectors are perpendicular to each other.

3e) for any non matrix B , if $B = PDP^{-1}$ where D is being the diagonal matrix $\mathbb{R}$

$$\therefore B^{-1} = PD^{-1}P^{-1}$$

~~for any~~ and $B = PDP^{-1}$

$$\therefore B^{-1} = (PDP^{-1})^{-1}$$

which can be written as $B^{-1} = (P^{-1})D^{-1}(P^{-1})^{-1}$

$$\boxed{B^{-1} = P^{-1}D^{-1}P}$$

$\therefore$ If $D = \text{diag}(D_{11}, D_{22}, \dots, D_{nn})$

$$D^{-1} = \text{diag}\left(\frac{1}{D_{11}}, \frac{1}{D_{22}}, \dots, \frac{1}{D_{nn}}\right)$$

$$\therefore D^{-1} = P^{-1}D^{-1}P$$

where D^{-1} is a diagonal matrix.

4. $u_t \rightarrow$ no. of people never been infected by day t
 $s_t \rightarrow$ no. of people sick on day t
 $r_t \rightarrow$ no. of people recovered by day t

o) for
$$\begin{pmatrix} u_{t+1} \\ s_{t+1} \\ r_{t+1} \end{pmatrix} = A \begin{pmatrix} u_t \\ s_t \\ r_t \end{pmatrix} \quad \text{find } A \text{ matrix}$$

$\therefore \Rightarrow$ for $u_{t+1} \therefore$ As 20% of u_t population gets sick

$$u_{t+1} = 0.8 u_t$$

$\Rightarrow$ for $s_{t+1} \therefore$ 20% of u_t people fall sick
 10% of r_t people get sick
 30% of s_t people remain sick

$$s_{t+1} = 0.2 u_t + 0.1 r_t + 0.3 s_t$$

$\Rightarrow$ for $r_{t+1} =$ 70% of s_t people recover
 90% of r_t people

$$r_{t+1} = 0.7 s_t + 0.9 r_t$$

$$\therefore \begin{pmatrix} u_{t+1} \\ s_{t+1} \\ r_{t+1} \end{pmatrix} = \begin{pmatrix} 0.8 & 0 & 0 \\ 0.2 & 0.3 & 0.1 \\ 0 & 0.7 & 0.9 \end{pmatrix} \begin{pmatrix} u_t \\ s_t \\ r_t \end{pmatrix}$$

$$\therefore A = \begin{bmatrix} 0.8 & 0 & 0 \\ 0.2 & 0.3 & 0.1 \\ 0 & 0.7 & 0.9 \end{bmatrix}$$

if $u_0 = 25$ (millions of people)

then $s_0 = 0$ and $r_0 = 0$

$\therefore$ to calculate $(u_2, s_2 \text{ and } r_2)$

$$\begin{bmatrix} u_1 \\ s_1 \\ r_1 \end{bmatrix} = \begin{bmatrix} 0.8 & 0 & 0 \\ 0.2 & 0.3 & 0.1 \\ 0 & 0.7 & 0.9 \end{bmatrix} \begin{bmatrix} 25 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 20 \\ 5 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} u_2 \\ s_2 \\ r_2 \end{bmatrix} = \begin{bmatrix} 0.8 & 0 & 0 \\ 0.2 & 0.3 & 0.1 \\ 0 & 0.7 & 0.9 \end{bmatrix} \begin{bmatrix} 20 \\ 5 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 16 \\ 5.5 \\ 3.5 \end{bmatrix}$$

$$\therefore (u_2, s_2, r_2) = (16, 5.5, 3.5)$$

6) if $v = (v_1, v_2)$

Show $\begin{pmatrix} 6t \\ t \\ -7t \end{pmatrix}$ is an eigenvector of A for any t

Using:- $Av(t) = \lambda v(t)$

$$v(t) = \begin{pmatrix} 6t \\ t \\ -7t \end{pmatrix}$$

$$\therefore Av(t) = \begin{bmatrix} 0.8 & 0.0 & 0.0 \\ 0.2 & 0.3 & 0.1 \\ 0.0 & 0.7 & 0.9 \end{bmatrix} \begin{bmatrix} 6t \\ t \\ -7t \end{bmatrix}$$

$$= \begin{bmatrix} 4.8t \\ 0.8t \\ -6.3t \end{bmatrix}$$

$$\Rightarrow 0.8 \begin{bmatrix} 6t \\ t \\ -7t \end{bmatrix}$$

$\therefore$ here $\lambda = 0.8$ which is the eigen value

c) If $\vec{v} = (v_1, v_2)$ is the eigenvector of matrix

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$$

then:-

$$C\vec{v} = \lambda \vec{v} \Rightarrow$$

$$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \lambda \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\begin{pmatrix} C_{11}v_1 + C_{12}v_2 \\ C_{21}v_1 + C_{22}v_2 \end{pmatrix} = \lambda \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \quad (1)$$

Using :- $B\vec{v}$

$$= \begin{pmatrix} b_1 & 0 & 0 \\ b_2 & C_{11} & C_{12} \\ b_3 & C_{21} & C_{22} \end{pmatrix} \begin{pmatrix} 0 \\ v_1 \\ v_2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 \\ C_{11}v_1 + C_{12}v_2 \\ C_{21}v_1 + C_{22}v_2 \end{pmatrix}$$

using (1) we can say that

$$B\vec{v} = \lambda \begin{pmatrix} 0 \\ v_1 \\ v_2 \end{pmatrix}$$

$$\therefore \boxed{B\vec{v} = \lambda \vec{v}}$$

$$d) \quad A = \begin{pmatrix} 0.8 & 0 & 0 \\ 0.2 & 0.3 & 0.1 \\ 0 & 0.7 & 0.9 \end{pmatrix}$$

$$\therefore \text{here } A - \lambda I = \begin{pmatrix} 0.8 & 0 & 0 \\ 0.2 & 0.3 & 0.1 \\ 0 & 0.7 & 0.9 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 0.8 - \lambda & 0 & 0 \\ 0.2 & 0.3 - \lambda & 0.1 \\ 0 & 0.7 & 0.9 - \lambda \end{pmatrix}$$

$$\therefore \det(A - \lambda I) = (0.8 - \lambda) [(0.3 - \lambda)(0.9 - \lambda)] - (0.1 \times 0.7)$$

$$= \lambda^2 - 1.2\lambda + 0.2$$

$$\therefore (\lambda^2 - 1.2\lambda + 0.2)$$

$$\Rightarrow \lambda^2 - 1\lambda - 0.2\lambda + 0.2$$

$$\Rightarrow \lambda(\lambda - 1) - 0.2(\lambda - 1)$$

$$\therefore (\lambda - 0.2)(\lambda - 1)$$

$$\therefore \boxed{\lambda = 0.2, 1}$$

~~$\therefore \text{for } \lambda = 1$~~

e) For Population after 365 days :-

$$A = PDP^{-1}$$

$$\therefore \text{for 365 days} \therefore A^{365} = P D^{365} P^{-1}$$

$$\therefore D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.2 & 0 \\ 0 & 0 & 0.8 \end{pmatrix}$$

$$\therefore D^{365} = \begin{pmatrix} 1^{365} & 0 & 0 \\ 0 & 0.2^{365} & 0 \\ 0 & 0 & 0.8^{365} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\therefore$ Approximately after 365 days (2 year) the no. of uninfected people will be 100% and sick and recovered will become zero.

4.

5. For the Set S we have three inequalities:- $T \cap U \cap P$

$$T \rightarrow (x-14)^2 + y^2 \leq 100$$

$$U \rightarrow x^2 + (y-14)^2 \leq 100$$

$$P \rightarrow x+y \geq 14$$

a) Describing the sets and their intersections:-

T:- It is a circle with center:- $(14, 0)$
radius:- 10

U:- It is a circle of center:- $(0, 14)$
radius:- 10

P:- It shows the region above the line made by $x+y=14$

$T \cap P$:- It shows the area which is inside the circle formed by T and above the line formed by P

$U \cap P$:- It shows the area inside the circle U which is above the line made by P.

$$\begin{array}{r} 196 \\ 96 \\ \hline 292 \end{array}$$

b) To find the intersection or the boundary points.
We have to solve the equations.

1. T and P

$$\Rightarrow (x-14)^2 + y^2 = 100 \quad \text{--- (1)}$$

$$\Rightarrow x+y=14 \quad \text{--- (2)}$$

$$\therefore x=14-y$$

$$\textcircled{1} \Rightarrow x^2 - 28x + 196 + y^2 = 100$$

$$x^2 + y^2 - 28x = -96$$

$$\textcircled{2} \Rightarrow x^2 + y^2 + 2xy = 196$$

$$\textcircled{2} - \textcircled{1} \Rightarrow 2xy + 28x = 292$$

$$\Rightarrow x(y+14) = 146$$

using value of x in (1)

$$(14-y-14)^2 + y^2 = 100$$

$$2y^2 = 100$$

$$y^2 = 50$$

$$y = \sqrt{50}$$

$$y = \pm 5\sqrt{2}$$

$$x_1 = 14 - 5\sqrt{2}$$

$$x_2 = 14 + 5\sqrt{2}$$

$\therefore$ the intersection points are: ~~$(5\sqrt{2}, 14-5\sqrt{2})$~~

$$(14-5\sqrt{2}, 5\sqrt{2})$$

$$(14+5\sqrt{2}, -5\sqrt{2})$$

2) V4P :-

$$x^2 + (y-14)^2 = 100$$

$$x + y = 14$$

$$y = 14 - x$$

$$x^2 + (14 - x - 14)^2 = 100$$

$$x^2 + x^2 = 100$$

$$2x^2 = 100$$

$$x^2 = 50$$

$$x = \pm 5\sqrt{2}$$

$$x_1 = -5\sqrt{2}$$

$$x_2 = +5\sqrt{2}$$

$$\therefore y_1 = 14 + 5\sqrt{2}$$

$$y_2 = 14 - 5\sqrt{2}$$

Intersection points :-
 $\therefore (-5\sqrt{2}, 14 + 5\sqrt{2})$
 $(5\sqrt{2}, 14 - 5\sqrt{2})$

2) TEU :-

$$(x-14)^2 + y^2 = 100 \quad \text{--- (1)} \quad x^2 - 28x + 196 + y^2 = 100$$

$$x^2 + (y-14)^2 = 100 \quad \text{--- (2)} \quad x^2 + y^2 - 28y + 196 = 100$$

$$-28x + 196 + 28y - 196 = 0$$

$$x - y = 0$$

$$x = y$$

$$\therefore 2x^2 - 28x = -96$$

$$2x(x-14) = -96$$

$$2x^2 - 28x + 96 = 0$$

$$x^2 - 14x + 48 = 0$$

$$(x-8)(x-6) = 0$$

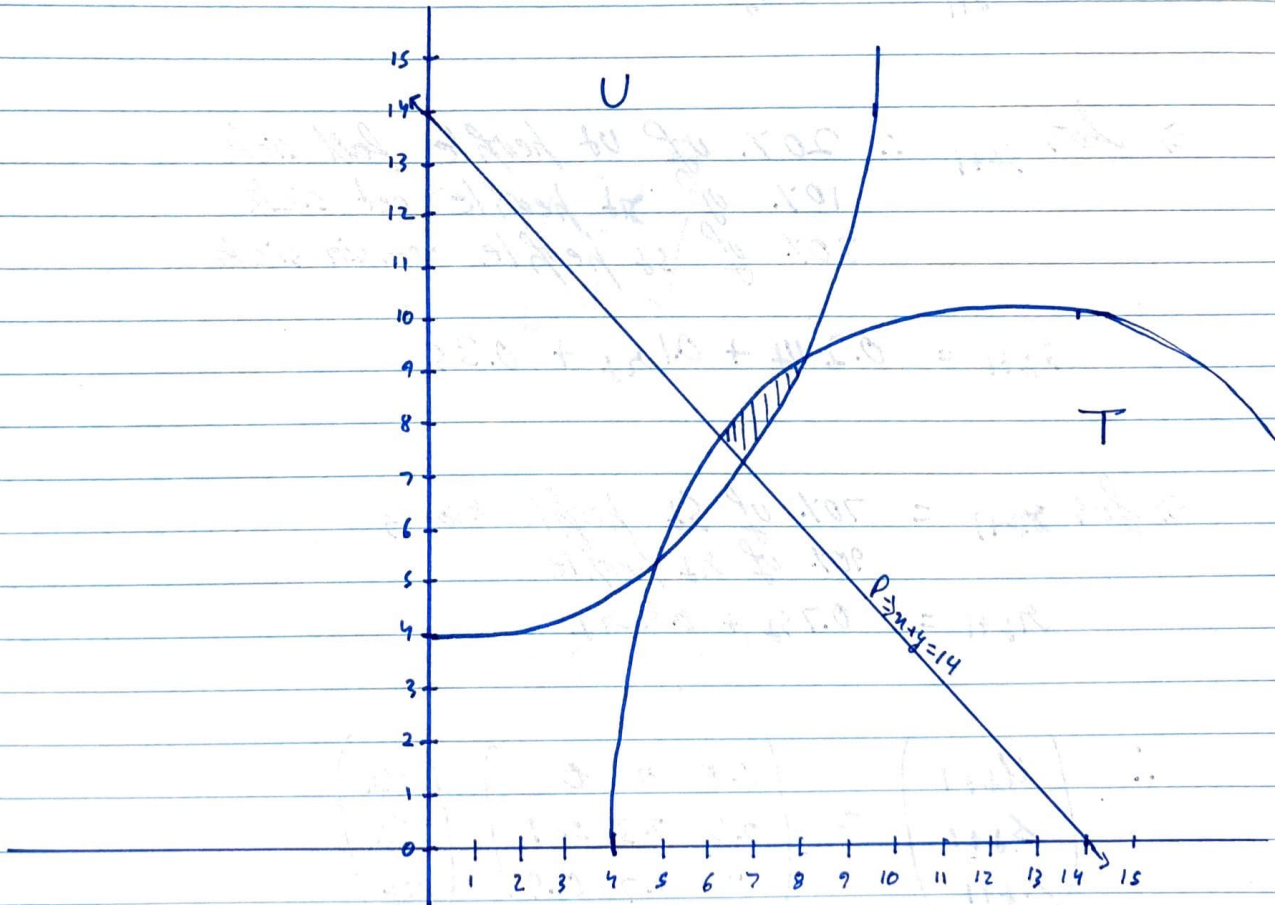
$$\therefore x_1 = 8$$

$$x_2 = 6$$

$\therefore$ the intersection points are:

$$(8, 8) \text{ \& } (6, 6)$$

ex



5c) To minimize $f(x, y) = y$ over the set S
where y is smallest.

$\therefore$ from the intersection pt. with the smallest
 y value is $(14 + 5\sqrt{2}, -5\sqrt{2})$

$\therefore$ Thus the pt. which minimises is $(14 + 5\sqrt{2}, -5\sqrt{2})$